

Fermionic Extensions of Continuous Functional Tensor Networks: Exchange-Rank and Exterior-Contraction Obstructions

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Hong et al. proposed continuous functional tensor networks by expanding each particle coordinate in an orthonormal functional basis, representing the coefficient tensor by a tensor network, evaluating basis-projected operators by contraction, and optimizing the tensors by automatic differentiation. We study the fermionic extension that their work left open. Two independent obstructions emerge. First, an ordinary particle-site tensor train pays an unavoidable binomial bond for exchange statistics; we recall the known flat particle spectrum of a single Slater determinant. Second, moving antisymmetry into exterior algebra removes that representation floor but does not make compact inputs easy to contract: exact squared-norm evaluation is $\#P$ -hard already at fixed one-form bond two under a polynomial-time metric reduction, and an independent reduction gives exact point-evaluation hardness in an explicit rational shifted-Legendre functional basis at the same bond. Along a matrix-pair subfamily, bounded and selected fixed-state memories collapse to polynomial linear combinations of established antisymmetrized geminal powers, while a bandwidth-one unique path already encodes a permanent. A generic relative squared-norm scheme would also approximate real positive-semidefinite permanents, whereas entrywise-nonnegative and additive promised cases remain open. Finally, cut-rank divisibility rejects a universal direct tensor product of a statistics carrier and a Slater-normalized correlation multiplicity. A nonbranching exterior subclass is evaluated as a numerical consequence of these constraints. Because that subclass is algebraically a finite nonorthogonal Slater expansion, the data are not claimed as a method distinct from nonorthogonal configuration interaction (NOCI). These exact obstructions constrain, but do not exclude, genuinely distinct restricted or controlled approximate algorithms; they do not establish a scalable generic fermionic solver.

Keywords: fermionic tensor networks, first quantization, exterior algebra, computational complexity, continuous functional basis

I. SCOPE AND THESIS

The direct motivation is the functional-tensor-network construction of Hong–Xiao–Hu–Ji–Ran [1]. For orthonormal local functions $\{\phi_s\}_{s=1}^D$, a first-quantized wavefunction is written

$$\Psi(x_1, \dots, x_N) = \sum_{s_1, \dots, s_N} C_{s_1 \dots s_N} \prod_{i=1}^N \phi_{s_i}(x_i). \quad (1)$$

Orthonormality turns the continuum norm into the Euclidean norm of C and turns differential and multiplicative operators into finite functional-basis matrices. Hong et al. compressed C by an MPS and optimized its Rayleigh quotient with automatic differentiation; they explicitly named fermionic tensor networks as a future electronic extension.

For identical fermions, however, $C \in \Lambda^N V_D$. An ordinary MPS over the labelled particle slots and an occupation-number MPS use different tensor product factorizations. Their bond dimensions are not related by the abstract Hilbert-space isomorphism. Hilbert-space MPS [2], Grassmann/graded networks [3], and the ordered first-quantized construction of Li–Waintal [4] are therefore comparators, not aliases for the representation studied here.

All complexity statements below use explicit finite encodings and specify the underlying field. Floating-point calculations are separated from the formal results and

are reported with their basis, truncation, optimization, and initialization controls. Antisymmetry is structural in every exterior or ordered-chamber construction below, so the corresponding residual is $\epsilon_{AS} = 0$. The generic exact-contraction obstruction motivates restricted and controlled approximate algorithms rather than excluding them.

The thesis is deliberately negative and representation-specific:

Ordinary particle TT stores exchange multiplicity, whereas compact exterior cores can hide hard coefficient evaluation. Removing the former cost does not remove the latter, and neither observation is a no-go theorem for every fermionic tensor network, ordered sector, determinant expansion, or promised approximation.

II. EXCHANGE RANK IN THE LABELLED-PARTICLE FACTORIZATION

Let V be a finite-dimensional real or complex Hilbert space. Identify $\Lambda^N V$ isometrically with the alternating subspace of $V^{\otimes N}$ by

$$J_N(v_1 \wedge \dots \wedge v_N) = \frac{1}{\sqrt{N!}} \sum_{\pi \in S_N} \text{sgn}(\pi) v_{\pi(1)} \otimes \dots \otimes v_{\pi(N)}.$$

Write $C_{(k)}$ for the $k \mid N - k$ particle matricization.

Theorem II.1 (Structural result I: exact particle-TT ranks). *For $1 \leq k < N$, the minimal exact ordinary-TT bond is*

$$r_k^{\text{TT}}(C) = \text{rank } C_{(k)}.$$

Proof. Fix an orthonormal basis $(e_s)_{s=1}^D$ of V and write $C_{s_1 \dots s_N}$ for the coefficients of C . An exact TT with bond χ_k at the $k | N - k$ cut has coefficients

$$C_{s_1 \dots s_N} = \sum_{\alpha_1=1}^{\chi_1} \dots \sum_{\alpha_{N-1}=1}^{\chi_{N-1}} G_{s_1 \alpha_1}^{[1]} G_{\alpha_1 s_2 \alpha_2}^{[2]} \dots G_{\alpha_{N-1} s_N}^{[N]}. \quad (2)$$

Contracting the cores on the two sides of the cut defines

$$(L_k)_{(s_1, \dots, s_k), \alpha_k} = \sum_{\alpha_1, \dots, \alpha_{k-1}} \prod_{i=1}^k G_{\alpha_{i-1} s_i \alpha_i}^{[i]},$$

$$(R_k)_{\alpha_k, (s_{k+1}, \dots, s_N)} = \sum_{\alpha_{k+1}, \dots, \alpha_{N-1}} \prod_{i=k+1}^N G_{\alpha_{i-1} s_i \alpha_i}^{[i]}, \quad (3)$$

where $\alpha_0 = \alpha_N = 1$. Equation (2) gives the entrywise identity

$$C_{(k)} = L_k R_k,$$

$$\text{rank } C_{(k)} \leq \min(\text{rank } L_k, \text{rank } R_k) \leq \chi_k. \quad (4)$$

For the converse, perform successive exact rank factorizations. At the first step write $C_{(1)} = L_1 Z_1$ with L_1 of full column rank $r_1 = \text{rank } C_{(1)}$. Inductively, suppose the first $k-1$ cores have been chosen so that their contraction L_{k-1} has full column rank and the remaining coefficient array is Z_{k-1} . Matricizing Z_{k-1} across its first virtual-plus-physical index gives a matrix $Z_{k-1}^{(k)}$ such that

$$C_{(k)} = (L_{k-1} \otimes I_D) Z_{k-1}^{(k)}. \quad (5)$$

Because $L_{k-1} \otimes I_D$ has full column rank, left multiplication by it is injective on columns, and hence

$$\text{rank } Z_{k-1}^{(k)} = \text{rank } C_{(k)} =: r_k. \quad (6)$$

Choose an exact rank factorization $Z_{k-1}^{(k)} = G^{[k]} Z_k$ of inner dimension r_k and reshape $G^{[k]}$ as the k th TT core. Continuing through $k = N-1$ produces one TT whose cut bonds are simultaneously $r_k = \text{rank } C_{(k)}$ and whose contracted coefficient tensor is C . This is the exact TT-SVD construction [5]. $\square$

Theorem II.2 (Structural result II: universal exchange floor). *If $0 \neq C \in \Lambda^N V$, then*

$$\text{rank } C_{(k)} \geq \binom{N}{k}.$$

The lower bound is attained by every nonzero decomposable N -form.

Proof. Expand

$$C = \sum_{I \subseteq [D], |I|=N} c_I e_I,$$

$$e_I = e_{i_1} \wedge \dots \wedge e_{i_N}, \quad i_1 < \dots < i_N. \quad (7)$$

and choose I with $c_I \neq 0$. Index rows of $C_{(k)}$ by ordered k -tuples and columns by ordered $(N-k)$ -tuples. For each k -subset $A \subset I$, take the increasing row tuple associated with A and the increasing column tuple associated with $I \setminus A$. Let M_I be the resulting $\binom{N}{k} \times \binom{N}{k}$ submatrix, with both its rows and columns indexed by k -subsets $A, B \subset I$ through

$$(M_I)_{A,B} := (C_{(k)})_{A, I \setminus B}. \quad (8)$$

If $A = B$, the concatenated indices are exactly I , so $(M_I)_{A,A} = \varepsilon_A c_I$ for a sign $\varepsilon_A \in \{\pm 1\}$. If $A \neq B$, equal cardinality implies $A \setminus B \neq \emptyset$. Every index in $A \setminus B$ appears both in A and in $I \setminus B$, so alternation gives $(M_I)_{A,B} = 0$. Consequently

$$\det M_I = \left(\prod_{A \subset I, |A|=k} \varepsilon_A \right) c_I^{\binom{N}{k}} \neq 0, \quad (9)$$

and therefore

$$\text{rank } C_{(k)} \geq \text{rank } M_I = \binom{N}{k}. \quad (10)$$

Now let $C = u_1 \wedge \dots \wedge u_N \neq 0$ and put $U = \text{span}\{u_1, \dots, u_N\}$, so $\dim U = N$. Every contraction of C by a k -covector lies in $\Lambda^{N-k} U$, whence

$$\text{rank } C_{(k)} \leq \dim \Lambda^{N-k} U = \binom{N}{N-k} = \binom{N}{k}. \quad (11)$$

Together with the lower bound, this proves equality for every nonzero decomposable form. $\square$

Corollary II.3 (strict-antisymmetry truncation floor). *If $\tilde{C} \neq 0$ and $r_k^{\text{TT}}(\tilde{C}) < \binom{N}{k}$ at any cut, then $\tilde{C} \notin \Lambda^N V$.*

Proof. If $\tilde{C}$ were in $\Lambda^N V$, Structural result II would give $\text{rank } \tilde{C}_{(k)} \geq \binom{N}{k}$. Structural result I gives the equality $r_k^{\text{TT}}(\tilde{C}) = \text{rank } \tilde{C}_{(k)}$, contradicting the assumed strict inequality. $\square$

The TT-rank characterization is standard. The exterior rank-floor proof is included for audit, but no priority claim is made pending a dedicated exterior-algebra literature review. It is distinct from the stronger, generally open entropy-minimization statements studied for fermionic reduced density matrices [4, 6].

Theorem II.4 (Structural result III: known flat Slater particle spectrum). *For orthonormal $u_1, \dots, u_N$, the*

normalized Slater determinant Φ has Schmidt decomposition

$$\Phi = \frac{1}{\sqrt{\binom{N}{k}}} \sum_{|I|=k} (-1)^{\epsilon(I)} \Phi_I \otimes \Phi_{I^c}.$$

Thus all $\binom{N}{k}$ nonzero singular values equal $\binom{N}{k}^{-1/2}$. The best cut-rank- r approximation has relative squared error

$$\epsilon_{k,r}^2 = 1 - \frac{r}{\binom{N}{k}},$$

so error at most ϵ requires $r_k^{\text{TT}} \geq \lceil (1 - \epsilon^2) \binom{N}{k} \rceil$.

Proof. Let $[N] = \{1, \dots, N\}$. For a subset $I = \{i_1 < \dots < i_k\} \subset [N]$, define the normalized antisymmetric tensors

$$\begin{aligned} \Phi_I &= \frac{1}{\sqrt{k!}} \sum_{\sigma \in S_k} \text{sgn}(\sigma) u_{i_{\sigma(1)}} \otimes \dots \otimes u_{i_{\sigma(k)}}, \\ \Phi_{I^c} &= \frac{1}{\sqrt{(N-k)!}} \sum_{\tau \in S_{N-k}} \text{sgn}(\tau) u_{j_{\tau(1)}} \otimes \dots \otimes u_{j_{\tau(N-k)}}, \end{aligned} \quad (12)$$

where $I^c = \{j_1 < \dots < j_{N-k}\}$. Let $\rho_I \in S_N$ be the shuffle that places $(i_1, \dots, i_k)$ in the first k positions and the complementary indices in the last $N-k$ positions, both in increasing order, and put $(-1)^{\epsilon(I)} = \text{sgn}(\rho_I)$.

Put $m := \binom{N}{k}$. Every permutation $\pi \in S_N$ has a unique decomposition into a subset I , the shuffle ρ_I , and internal permutations $(\sigma, \tau) \in S_k \times S_{N-k}$. Substitution in the normalized Slater sum therefore gives

$$\begin{aligned} \Phi &= \frac{1}{\sqrt{N!}} \sum_{|I|=k} (-1)^{\epsilon(I)} (\sqrt{k!} \Phi_I) \otimes (\sqrt{(N-k)!} \Phi_{I^c}) \\ &= \sqrt{\frac{k!(N-k)!}{N!}} \\ &\quad \times \sum_{|I|=k} (-1)^{\epsilon(I)} \Phi_I \otimes \Phi_{I^c} \\ &= m^{-1/2} \\ &\quad \times \sum_{|I|=k} (-1)^{\epsilon(I)} \Phi_I \otimes \Phi_{I^c}. \end{aligned} \quad (13)$$

Orthonormality of the u_i implies $\langle \Phi_I, \Phi_J \rangle = \delta_{IJ}$ and likewise for the complements. The displayed expression is therefore a Schmidt decomposition with exactly m singular values $s_j = m^{-1/2}$. By Eckart–Young,

$$\min_{\text{rank } X \leq r} \frac{\|\Phi - X\|_2^2}{\|\Phi\|_2^2} = \sum_{j=r+1}^m s_j^2 = \frac{m-r}{m} = 1 - \frac{r}{\binom{N}{k}}. \quad (14)$$

Thus squared error at most ϵ^2 requires $r \geq (1 - \epsilon^2) \binom{N}{k}$, and integrality gives the stated ceiling. This is the standard particle-reduced-density-matrix structure of a Slater determinant [7]. $\square$

No flat spectrum is asserted for a general correlated alternating tensor. Theorems II.1, II.2, and II.4 concern labelled-particle TT only; they do not apply to mode MPS, ordered chambers, or exterior/determinant carriers.

For a fixed orthonormal one-particle basis of size D , all of these representations describe states in exactly the same finite physical space $\Lambda^N V_D$, whose full-configuration-interaction dimension is $\binom{D}{N}$. A mode-space FCI or quantum-chemistry DMRG calculation in that orbital basis changes the factorization and variational compression, not the physical finite-basis space [2, 8]. Consequently, FEMPS has neither a larger variational space nor an accuracy or efficiency advantage merely by being first quantized. Such an advantage would require matched basis, tolerance, timing, and memory comparisons.

III. EXTERIOR MATRIX-PRODUCT REPRESENTATION

Definition III.1 (one-form matrix-wedge state). For site j , let $A^{[j]} \in \text{Mat}_{\chi_{j-1} \times \chi_j}(V_D)$ with $\chi_0 = \chi_N = 1$. For compatible matrices of forms define

$$(X \wedge Y)_{ac} = \sum_b X_{ab} \wedge Y_{bc}.$$

The open-boundary state is

$$C(A^{[1]}, \dots, A^{[N]}) = [A^{[1]} \wedge \dots \wedge A^{[N]}]_{11} \in \Lambda^N V_D. \quad (15)$$

Exterior associativity makes matrix-wedge multiplication associative, and each virtual path is a wedge of N one-forms. Hence the state is alternating by construction and $\epsilon_{\text{AS}} = 0$. The $\chi = 1$ family is exactly the decomposable forms. A sum of R Slaters embeds by R diagonal virtual paths, and the ordinary scalar MPS gauge action remains valid. These elementary facts do not supply a canonical form or an efficient norm contraction. The displayed parameter count $O(ND\chi^2)$ is an input-size statement only.

IV. FIXED-BOND EXACT CONTRACTION OBSTRUCTION

For an $n \times n$ matrix $M = (M_{ij})$ over a noncommutative algebra, define the row-ordered Cayley determinant

$$\text{CDet}(M) = \sum_{\sigma \in S_n} \text{sgn}(\sigma) M_{1,\sigma(1)} \cdots M_{n,\sigma(n)}.$$

Chien–Harsha–Sinclair–Srinivasan prove #P-hardness over $\text{Mat}_2(\mathbb{F})$ in characteristic zero [9]; related hardness results for the noncommutative determinant are given by Arvind–Srinivasan [10].

The distinction from exchange is essential. For scalar entries ($\chi = 1$), the factors commute, so the same alternating permutation sum is the ordinary determinant and is computable by Gaussian elimination; in FEMPS it is just the Slater exchange carrier. At $\chi = 2$, the entries lie in the fixed four-dimensional algebra $\text{Mat}_2(\mathbb{Q})$ and the product must remain in row order. The final physical wavefunction is still scalar and exactly antisymmetric: the noncommutativity lives only in the virtual computation.

Lemma IV.1 (direct Cayley coefficient). *Let $M_{ij} \in \text{Mat}_d(\mathbb{F})$ and set $\mathcal{A}^{[i]} = \sum_j M_{ij} e_j$. For $u, v \in \mathbb{F}^d$,*

$$u^\top \mathcal{A}^{[1]} \wedge \cdots \wedge \mathcal{A}^{[n]} v = (u^\top \text{CDet}(M)v) e_1 \wedge \cdots \wedge e_n.$$

After absorbing the boundaries into the endpoint cores, this is a FEMPS with $D = N = n$ and maximum internal bond d .

Proof. By distributivity of matrix multiplication and the exterior product,

$$\begin{aligned} u^\top \mathcal{A}^{[1]} \wedge \cdots \wedge \mathcal{A}^{[n]} v &= \sum_{j_1, \dots, j_n=1}^n u^\top M_{1j_1} \cdots M_{nj_n} v \\ &\quad \times e_{j_1} \wedge \cdots \wedge e_{j_n}. \end{aligned} \quad (16)$$

If two indices j_a, j_b coincide, the corresponding exterior monomial is zero. Every nonzero summand therefore has $(j_1, \dots, j_n) = (\sigma(1), \dots, \sigma(n))$ for a unique $\sigma \in S_n$, and

$$e_{\sigma(1)} \wedge \cdots \wedge e_{\sigma(n)} = \text{sgn}(\sigma) e_1 \wedge \cdots \wedge e_n. \quad (17)$$

Substitution into Eq. (16) yields

$$\begin{aligned} u^\top \mathcal{A}^{[1]} \wedge \cdots \wedge \mathcal{A}^{[n]} v &= \sum_{\sigma \in S_n} \text{sgn}(\sigma) u^\top M_{1\sigma(1)} \cdots M_{n\sigma(n)} v \\ &\quad \times e_1 \wedge \cdots \wedge e_n \\ &= (u^\top \text{CDet}(M)v) e_1 \wedge \cdots \wedge e_n. \end{aligned} \quad (18)$$

Replacing the first matrix core by $u^\top \mathcal{A}^{[1]}$ and the last by $\mathcal{A}^{[n]} v$ changes the endpoint dimensions to one and leaves all internal dimensions at most d , proving the final sentence of the lemma. $\square$

Theorem IV.2 (Fixed-bond squared-norm hardness). *Over $\mathbb{Q}$, exact squared-norm evaluation for rational one-form FEMPS is $\#P$ -hard under a polynomial-time metric reduction even when $D = N$ and every internal bond is at most two. A generic exact contraction algorithm for fixed $\chi = 2$ would imply $\text{FP} = \#P$.*

Proof. Source instance. Let φ be a 3CNF formula with m clauses, n_{var} variables, and

$$S := \#\{z \in \{0, 1\}^{n_{\text{var}}} : \varphi(z) = 1\}. \quad (19)$$

Theorems 3.5 and 3.9 of Chien–Harsha–Sinclair–Srinivasan (CHSS) construct from φ in polynomial time an $n \times n$ matrix

$$H_\varphi = (H_{ij})_{i,j \in [n]}, \quad H_{ij} \in \text{Mat}_2(\mathbb{Q}), \quad (20)$$

where $n = \text{poly}(|\varphi|)$ and the gadget entries are constant-bit integers, such that

$$\begin{aligned} \text{CDet}(H_\varphi) &= aI_2 + bJ_2, \quad J_2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \\ a + b &= 4^{3m} S. \end{aligned} \quad (21)$$

CHSS label the graph vertices, and their determinant is the row-ordered Cayley determinant in that vertex order.

AUDIT FLAG. The structured identity $\text{CDet}(H_\varphi) = aI_2 + bJ_2$ with $a + b = 4^{3m} S$, including the polynomial construction, row ordering, and constant-bit gadget claim, is imported from CHSS Theorems 3.5 and 3.9. It is not derived from the preceding FEMPS definitions and requires independent source verification. The standard $\#P$ -completeness of $\#SAT$ is also used as the source-problem classification rather than reproved here.

Constructed FEMPS instance. For $i \in [n]$ define the one-form matrix core

$$\mathcal{A}^{[i]} := \sum_{j=1}^n H_{ij} e_j \in \text{Mat}_2(V_n), \quad (22)$$

and choose $u = e_1^{(2)} = (1, 0)^\top$ and $v = e_1^{(2)} + e_2^{(2)} = (1, 1)^\top$. Absorb $u^\top$ and v into the endpoint cores. The constructed instance has

$$N = D = n, \quad \chi_i = 2 \quad (1 \leq i < n). \quad (23)$$

By the direct Cayley coefficient lemma its state is

$$C_\varphi = c_\varphi e_1 \wedge \cdots \wedge e_n, \quad c_\varphi = u^\top \text{CDet}(H_\varphi) v. \quad (24)$$

The boundary multiplication is explicit:

$$\begin{aligned} c_\varphi &= (1, 0)(aI_2 + bJ_2)(1, 1)^\top \\ &= (1, 0) \begin{pmatrix} a & b \\ b & a \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = a + b = 4^{3m} S \geq 0. \end{aligned} \quad (25)$$

Oracle value. The physical basis is orthonormal, so $\|e_1 \wedge \cdots \wedge e_n\|_2 = 1$. An exact squared-norm oracle therefore returns

$$y_\varphi := \|C_\varphi\|_2^2 = c_\varphi^2 = 16^{3m} S^2. \quad (26)$$

Source answer and postprocessing. Because Eq. (25) is nonnegative, the integer square root is unambiguous and

$$S = \frac{\sqrt{y_\varphi}}{4^{3m}}. \quad (27)$$

Exact integer square root and division are polynomial-time in the output bit length.

Size and bit audit. There are n^2 matrices H_{ij} , each with four constant-bit entries, so the constructed FEMPS coefficient list has $O(n^2)$ rational scalars and polynomial encoding length. The bond is exactly bounded by two, independently of n . Moreover $0 \leq S \leq 2^{n_{\text{var}}}$, hence

$$\text{bits}(c_\varphi) \leq 6m + n_{\text{var}} + O(1), \quad (28)$$

$$\text{bits}(y_\varphi) \leq 12m + 2n_{\text{var}} + O(1), \quad (29)$$

which are polynomial in $|\varphi|$. Thus the complete chain

$$\varphi \mapsto (\mathcal{A}^{[1]}, \dots, \mathcal{A}^{[n]}) \mapsto y_\varphi \mapsto \frac{\sqrt{y_\varphi}}{4^{3m}} = S \quad (30)$$

is a polynomial-time metric reduction. $\square$

The structured CHSS boundary is what sharpens the bond from three to two. For an arbitrary signed Cayley entry x , one may still adjoin the bond-one reference $\Theta = e_1 \wedge \dots \wedge e_n$ by a block direct sum. The two norms x^2 and $(x+1)^2$ recover $x = ((x+1)^2 - x^2 - 1)/2$ at maximum bond three. That polarization observation is useful for general outputs, but it is not needed for the CHSS reduction.

CHSS work over any field of characteristic $p \neq 2$: their characteristic-zero statement is $\#P$ -hard and their positive odd-characteristic statement is Mod_pP -hard. We specialize to $\mathbb{Q}$ so that the input, output, square root, and bit-complexity model are explicit; no statement is made in characteristic two.

The result is not in conflict with the polynomial collapse of a fixed Mat_2 homogeneous pair power below. The hard one-form cores are site-dependent and preserve row order. Repeating one even two-form instead symmetrizes factor order.

The direct coefficient identity is pointwise algebraic and also holds when the entries are one-particle functions evaluated at a coordinate. A separate reduction for an explicit rational functional basis is proved in Theorem X.1; it is independent of the norm reduction even though both start from the same CHSS source. Neither result says that every $\chi \geq 2$ FEMPS is unsuitable for Monte Carlo or controlled approximation.

A. Why parity is only part of the mechanism

The contrast above is not an odd-versus-even dichotomy. Site-dependent odd one-form cores retain the ordered virtual product and can encode the hard row-ordered determinant. Repetition of one homogeneous even two-form makes the physical factors commute and can collapse a fixed coefficient algebra. But scalar-bond odd one-form states are just Slater determinants and are easy, whereas the growing-width even sparse-path family below is hard. Parity explains the commutation step, while site dependence, homogeneity, virtual width, and algebraic closure decide the contraction problem. Splitting a general matrix-valued pair core into consecutive one-form cores can also increase the bond by a factor of order D ; it is not a cost-free equivalence.

B. Independent tagged pair-power mechanism

For $N = 2M$, a matrix-valued two-form $\Omega_B = \sum_{p < q} B_{pq} e_p \wedge e_q$ defines

$$\Psi_M(B; l, r) = l^\top \frac{\Omega_B^M}{M!} r.$$

Shift tags $E_{i,i+1} \otimes M_{ij}$ annihilate all factor orders except increasing row order. This gives a separate polynomial-time Turing reduction with pair bond $2(n+1)$ and one-form bond $O(N^2)$. It is weaker than Theorem IV.2 for unrestricted one-form cores, but isolates growing non-commutative order memory inside the homogeneous pair-power subfamily.

V. THE MATRIX-PAIR CORRIDOR

A. Bounded Wedderburn–radical memory

Theorem V.1 (Bounded-algebra LC–AGP collapse). *Let $\mathcal{A}$ be a finite-dimensional unital complex algebra with radical $R^d = 0$ and Wedderburn–Malcev decomposition*

$$\mathcal{A} = S \oplus R, \quad S = \bigoplus_{a=1}^q \text{Mat}_{p_a}(\mathbb{C}).$$

Put $p = \max p_a$ and $\rho = \dim R$. For any linear boundary functional λ , $\lambda(\Omega^M)/M!$ is an exact LC–AGP with

$$K \leq \sum_{k=0}^{\min(d-1, M)} q^{k+1} \rho^k \binom{M + v_k - 1}{v_k - 1}, \quad (31)$$

$$v_k = (k+1)p^2 + k.$$

For fixed p and d , this is polynomial in the explicit input dimensions. If all data, the displayed decomposition, and the boundary are supplied over $\mathbb{Q}$ with polynomial-bit structure constants, the LC–AGP coefficients can be constructed in deterministic polynomial Turing time and have polynomial bit length. No algorithm for discovering a rational Wedderburn–Malcev decomposition is asserted here.

Proof. Let $E = \Lambda^{\text{even}} V_D$ and $E_2 = \Lambda^2 V_D$. The physical coefficients lie in the commutative algebra E , and

$$\Omega = \Omega_S + \Omega_R, \quad \Omega_S \in S \otimes E_2, \quad \Omega_R \in R \otimes E_2. \quad (32)$$

Expand Ω^M as a sum of length- M words in the two symbols Ω_S, Ω_R . If a word contains k radical factors, the two-sided ideal identities

$$SR^j \subseteq R^j, \quad R^j S \subseteq R^j, \quad R^i R^j \subseteq R^{i+j} \quad (33)$$

place its virtual coefficient in R^k . It vanishes for $k \geq d$. Hence

$$\lambda(\Omega^M) = \sum_{k=0}^{\min(d-1, M)} \lambda(W_{M,k}), \quad (34)$$

where $W_{M,k}$ is the sum of words with exactly k radical factors.

Fix such a k . Every word has a unique run-length vector $(m_0, \dots, m_k) \in \mathbb{Z}_{\geq 0}^{k+1}$ with $m_0 + \dots + m_k = M - k$ and virtual order

$$\Omega_S^{m_0} \Omega_R \Omega_S^{m_1} \dots \Omega_R \Omega_S^{m_k}. \quad (35)$$

Insert the q central block identities independently in the $k + 1$ semisimple runs and expand each radical occurrence in a fixed basis $(r_b)_{b=1}^\rho$ of R . A structural label is therefore

$$\tau = (a_0, b_1, a_1, \dots, b_k, a_k) \in [q]^{k+1} \times [\rho]^k, \quad (36)$$

so there are at most $q^{k+1} \rho^k$ labels.

For fixed τ , express every semisimple block in matrix units. The j th run uses at most $p_{a_j}^2 \leq p^2$ scalar two-forms, and the k selected radical factors contribute at most k additional scalar two-forms. Thus, as shown term by term in Appendix A, the boundary value is a homogeneous degree- M polynomial

$$P_\tau(z_1, \dots, z_{v(\tau)}) \in E_{2M}, \quad v(\tau) \leq (k+1)p^2 + k = v_k. \quad (37)$$

The variables commute because they are physical two-forms. Over characteristic zero, the degree- M Veronese powers span the homogeneous polynomials in v_k variables. Appendix A supplies an explicit unisolvent integer simplex of cardinality

$$s_k = \#\{\alpha \in \mathbb{Z}_{\geq 0}^{v_k} : |\alpha| = M\} = \binom{M + v_k - 1}{v_k - 1}, \quad (38)$$

so P_τ is a linear combination of at most s_k powers L^M . After division by $M!$, every $L^M/M!$ is one AGP. Multiplying s_k by the number of structural labels and summing Eq. (34) gives

$$K \leq \sum_{k=0}^{\min(d-1, M)} q^{k+1} \rho^k \binom{M + v_k - 1}{v_k - 1}, \quad (39)$$

which is Eq. (31).

For the rational Turing statement, p and d are fixed, so each v_k and the largest exponent k are fixed. Consequently

$$q^{k+1} \rho^k = \text{poly}(q, \rho), \quad s_k = O(M^{v_k-1}). \quad (40)$$

With the rational block decomposition, radical basis, and boundary supplied, all structure constants entering Eq. (37) have polynomial bit length. The integer-simplex evaluation matrix has polynomial order and polynomial-bit entries; the Hadamard and exact-elimination bounds in Appendix A give polynomial-bit inverse coefficients. Thus the number of arithmetic operations and every produced coefficient bit length are polynomial in the explicit input size. No step discovers a Wedderburn–Malcev decomposition. $\square$

For full Mat_2 , the bound specializes to at most $\binom{M+3}{3}$ AGPs even though the row-ordered determinant over the same matrix algebra is hard. This theorem classifies a pair-power family; it does not give an algorithm for the site-dependent cores of Theorem IV.2.

B. Selected growing memories

Theorem V.2 (Fixed-state graded LC–AGP collapse). *Suppose the coefficient algebra embeds faithfully in*

$$\text{Mat}_w \left(\mathbb{C}[z_1, \dots, z_g] / (z_1^{d_1}, \dots, z_g^{d_g}) \right),$$

with fixed w, g . Then every boundary contraction of $\Omega^M/M!$ is an exact LC–AGP with

$$K \leq \prod_{j=1}^g [M(d_j - 1) + 1] \binom{M + w^2 - 1}{w^2 - 1}.$$

In particular, the one-generator algebra $\mathbb{C}[z]/z^d$ requires at most $M(d - 1) + 1$ AGPs. For rational inputs with the embedding and quotient basis supplied explicitly, fixed w, g also gives a deterministic polynomial-time exact construction with polynomial-bit coefficients.

Proof. Write

$$R_d = \mathbb{C}[z_1, \dots, z_g] / (z_1^{d_1}, \dots, z_g^{d_g}) \quad (41)$$

and use the supplied embedding to regard Ω as a $w \times w$ matrix whose entries are two-forms with coefficients in R_d . Choose for every residue class its canonical lift with $0 \leq \deg_{z_j} < d_j$. In an unreduced product of M lifted entries,

$$\deg_{z_j} \leq L_j := M(d_j - 1) \quad (1 \leq j \leq g). \quad (42)$$

For counter z_j , evaluate on the $L_j + 1$ distinct nodes $T_j = \{0, 1, \dots, L_j\}$. The associated Vandermonde matrix

$$(V_j)_{t,m} = t^m, \quad t, m \in \{0, \dots, L_j\}, \quad (43)$$

is nonsingular. Hence the tensor-product matrix $V_1 \otimes \dots \otimes V_g$ recovers every coefficient of every lifted matrix entry from

$$G := \prod_{j=1}^g (L_j + 1) = \prod_{j=1}^g [M(d_j - 1) + 1] \quad (44)$$

grid evaluations. Reducing the recovered polynomial modulo $z_j^{d_j}$ only discards monomials and cannot increase the number of terms.

At a fixed grid point $t = (t_1, \dots, t_g)$, denote the w^2 scalar two-form entries of the evaluated matrix by $x_{ab}(t)$, $a, b \in [w]$. The boundary contraction of its M th matrix power is a homogeneous degree- M polynomial in those

w^2 commuting variables. The same integer-simplex polarization used in Appendix A expresses this polynomial using at most

$$S = \binom{M + w^2 - 1}{w^2 - 1} \quad (45)$$

powers of scalar two-forms. Interpolating the G grid values therefore uses at most

$$K \leq GS = \text{prod}_{j=1}^g [M(d_j - 1) + 1] \binom{M + w^2 - 1}{w^2 - 1}, \quad (46)$$

which is Eq. (V.2).

For rational input, all nodes are integers. Appendix B uses Hadamard bounds and exact Gaussian elimination to show that the inverse Vandermonde entries have polynomial bit length. With fixed g ,

$$G \leq M^g \prod_{j=1}^g \max(1, d_j) \quad (47)$$

up to a fixed g -dependent factor, and with fixed w , $S = O(M^{w^2-1})$. Because the quotient basis has explicit dimension $\prod_j d_j$, the grid, matrix multiplications, interpolation, quotient reduction, boundary application, and division by $M!$ all require polynomial time and produce polynomial-bit coefficients. This construction uses the supplied embedding and does not discover it. $\square$

The theorem includes a noncommutative alternating-word algebra of dimension $2d - 1$ embedded in $\text{Mat}_2(\mathbb{C}[z]/z^d)$. Thus growing nilpotency and two path branches remain insufficient when automaton width and counter count are fixed. Weighted-automata and Waring/Veronese ingredients are established prior art; the project-specific result is only their LC-AGP classification.

C. A bandwidth-one hard path

Let $P_j = e_{2j-1} \wedge e_{2j}$ be orthonormal pair forms. Given $A \in \mathbb{F}^{M \times M}$, define edge forms $F_i = \sum_j A_{ij} P_j$ on the unique endpoint path of an upper-bidiagonal $(M+1) \times (M+1)$ pair matrix.

Theorem V.3 (Sparse APG permanent obstruction). *Over $\mathbb{Q}$ (or $\mathbb{R}$ with its positive inner product), for a zero-one matrix A the endpoint state and its squared norm are*

$$\Psi_A = \frac{F_1 \wedge \cdots \wedge F_M}{M!} = \frac{\text{perm}(A)}{M!} P_1 \wedge \cdots \wedge P_M, \quad (48)$$

$$\langle \Psi_A, \Psi_A \rangle = \frac{\text{perm}(A)^2}{(M!)^2}.$$

Exact squared-norm evaluation is therefore #P-hard for a family with pair bandwidth one, one virtual path, $D = 2M$, and $O(M^2)$ binary data. Over $\mathbb{C}$, the norm formula is $|\text{perm}(A)|^2 / (M!)^2$ and the same zero-one reduction applies.

Proof. Source instance and constructed state. Let $A = (A_{ij})_{i,j \in [M]} \in \{0,1\}^{M \times M}$ be a permanent instance and set

$$P_j = e_{2j-1} \wedge e_{2j}, \quad F_i = \sum_{j=1}^M A_{ij} P_j. \quad (49)$$

Place F_i in the $(i, i+1)$ entry of an $(M+1) \times (M+1)$ upper-bidiagonal pair matrix and set every other entry to zero. There is exactly one virtual path from endpoint 1 to endpoint $M+1$, so its endpoint pair-power state is

$$\Psi_A = \frac{1}{M!} F_1 \wedge \cdots \wedge F_M. \quad (50)$$

Expanding the right-hand side gives

$$F_1 \wedge \cdots \wedge F_M = \sum_{j_1, \dots, j_M \in [M]} \left(\prod_{i=1}^M A_{i, j_i} \right) P_{j_1} \wedge \cdots \wedge P_{j_M}. \quad (51)$$

Because $P_j \wedge P_j = 0$, a summand is nonzero only if $i \mapsto j_i$ is injective, hence a permutation $\sigma \in S_M$. Since $\deg P_j = 2$, graded commutativity gives $P_i \wedge P_j = (-1)^{2 \cdot 2} P_j \wedge P_i = P_j \wedge P_i$. Thus every surviving term has the same ordered physical factor and

$$F_1 \wedge \cdots \wedge F_M = \sum_{\sigma \in S_M} \left(\prod_{i=1}^M A_{i, \sigma(i)} \right) P_1 \wedge \cdots \wedge P_M = \text{perm}(A) P_1 \wedge \cdots \wedge P_M. \quad (52)$$

Substitution in Eq. (50) proves the state formula.

Oracle value and recovery. The P_j are orthonormal and use disjoint one-particle basis vectors, so $\|P_1 \wedge \cdots \wedge P_M\|_2 = 1$. Hence an exact squared-norm oracle returns

$$y_A := \langle \Psi_A, \Psi_A \rangle = \frac{\text{perm}(A)^2}{(M!)^2}. \quad (53)$$

Since A is zero-one, $\text{perm}(A) \in \mathbb{Z}_{\geq 0}$ and

$$\text{perm}(A) = \sqrt{(M!)^2 y_A}. \quad (54)$$

Size, bond, and postprocessing audit. The physical dimension is $D = 2M$, the virtual width is $M+1$, and the upper-bidiagonal matrix contains exactly M nonzero two-form entries described by the M^2 input bits of A . The one-path property does not enumerate permutations. Finally, $\log_2(M!) = O(M \log M)$ and $0 \leq \text{perm}(A) \leq M!$, so exact multiplication and non-negative integer square root take polynomial time in the source and oracle-output lengths. The chain

$$A \mapsto (F_1, \dots, F_M) \mapsto y_A \mapsto \sqrt{(M!)^2 y_A} = \text{perm}(A) \quad (55)$$

is therefore a polynomial-time metric reduction, and Valiant's theorem applies [11]. Over $\mathbb{C}$, the same calculation replaces the square by the absolute square; the zero-one source keeps the recovered value nonnegative.

AUDIT FLAG. The final #P-hardness conclusion imports Valiant's theorem that the permanent of a zero-one matrix is #P-hard. That source theorem is cited but not reproved in this manuscript [11]. $\square$

The state is an established antisymmetrized product of geminals, and permanent-valued APG/APIG coefficients and APG-to-AGP decompositions have direct prior art [12, 13]. The hard output is projectively a single top-form Slater, so this is an input-to-coefficient obstruction, not a lower bound on physical LC-AGP rank.

VI. RELATIVE APPROXIMATION AND ENERGY CERTIFICATION

Exact permanent hardness alone does not exclude approximation. In particular, entrywise-nonnegative permanents admit an FPRAS [14]. The same sparse-path identity also accepts a different promise.

Theorem VI.1 (Real-PSD relative-norm transfer). *Unless $\text{RP} = \text{NP}$, there is no randomized polynomial-time relative approximation scheme for the squared norm of every rational real upper-bidiagonal matrix-pair input, polynomial jointly in dimensions, input bit length, inverse tolerance, and logarithmic inverse failure probability.*

Proof. Source instance and construction. Let $A \in \mathbb{Q}^{M \times M}$ be real positive semidefinite and let B be the maximum bit length of its rational entries. Apply the sparse-path map from the preceding theorem. It produces $D = 2M$, virtual width $M+1$, and $O(M^2)$ rational coefficients of bit length $O(B)$. Its exact squared norm is

$$s_A = \frac{\text{perm}(A)^2}{(M!)^2}, \quad \text{perm}(A) \geq 0. \quad (56)$$

Oracle value. Assume the squared-norm scheme in the theorem statement exists. On input tolerance $0 < \eta < 1$ and failure probability $0 < \delta < 1$, it returns $\tilde{s}_A$ such that, with probability at least $1 - \delta$,

$$|\tilde{s}_A - s_A| \leq \eta s_A. \quad (57)$$

On this event, if $s_A > 0$ then

$$\left| \frac{\sqrt{\tilde{s}_A}}{\sqrt{s_A}} - 1 \right| = \frac{|\tilde{s}_A/s_A - 1|}{\sqrt{\tilde{s}_A/s_A + 1}} \leq \eta. \quad (58)$$

If $s_A = 0$, Eq. (57) forces $\tilde{s}_A = 0$, so the same recovery is exact.

Source answer and complexity. Define

$$\tilde{p}_A := M! \sqrt{\tilde{s}_A}. \quad (59)$$

Equations (56) and (58) give

$$|\tilde{p}_A - \text{perm}(A)| \leq \eta \text{perm}(A) \quad (60)$$

on the same success event. Computing $M!$ and the square root to the requested relative precision uses time polynomial in M , B , and $\log(1/\eta)$; the assumed oracle is polynomial in the weaker parameters listed in the theorem, so this is a randomized polynomial-time relative approximation for the real-PSD permanent. The reduction chain is

$$A \mapsto (F_1, \dots, F_M) \mapsto \tilde{s}_A \mapsto M! \sqrt{\tilde{s}_A} = \tilde{p}_A. \quad (61)$$

AUDIT FLAG. The final contradiction uses the precise approximation theorem attributed to Meiburg: a PRAS for rational real positive-semidefinite permanents would imply $\text{RP} = \text{NP}$. That source theorem, including its input promise and approximation convention, is not reproved here and requires independent source verification [15]. $\square$

Additive permanent estimators remain possible [16]. They do not uniformly imply relative norm control: cancellations can make the true coefficient exponentially small in input precision while the additive scale remains order one.

Lemma VI.2 (Rayleigh-quotient certificate). *If $n > 0$, $E = h/n$, and estimates obey*

$$|n - \tilde{n}| \leq \Delta_n, \quad |h - \tilde{h}| \leq \Delta_h, \quad \tilde{n} > \Delta_n,$$

then, with $\tilde{E} = \tilde{h}/\tilde{n}$,

$$|E - \tilde{E}| \leq \frac{\Delta_h + |\tilde{E}| \Delta_n}{\tilde{n} - \Delta_n}.$$

For random bounds this holds on the simultaneous success event.

Proof. The norm estimate gives the explicit lower bound

$$n \geq \tilde{n} - |n - \tilde{n}| \geq \tilde{n} - \Delta_n > 0. \quad (62)$$

Using $h = nE$ and $\tilde{h} = \tilde{n}\tilde{E}$,

$$\begin{aligned} n(E - \tilde{E}) &= h - n\tilde{E} \\ &= (h - \tilde{h}) + \tilde{h} - n\tilde{E} \\ &= (h - \tilde{h}) - \tilde{E}(n - \tilde{n}). \end{aligned} \quad (63)$$

The triangle inequality and the assumed error bounds therefore imply

$$\begin{aligned} |E - \tilde{E}| &\leq \frac{|h - \tilde{h}| + |\tilde{E}| |n - \tilde{n}|}{n} \\ &\leq \frac{\Delta_h + |\tilde{E}| \Delta_n}{\tilde{n} - \Delta_n}, \end{aligned} \quad (64)$$

where Eq. (62) was used in the last step. If the bounds are random, the same deterministic derivation holds on their simultaneous success event. $\square$

Thus estimator unbiasedness alone is not an energy certificate; the denominator interval must stay positive. Tensor-network Monte Carlo [17] and sign-problem complexity [18] provide context but are not used as the reduction.

VII. FAILURE OF A UNIVERSAL DIRECT STATISTICS CARRIER

For $0 \neq C \in \Lambda^N V$, define the cut contraction

$$c_{C,k} : \Lambda^k V^* \longrightarrow \Lambda^{N-k} V, \quad c_{C,k}(\alpha) = \iota_\alpha C,$$

and $\mathcal{B}_k(C) = \text{im } c_{C,k}$. The tested proposal was an exact universal factorization

$$\mathcal{B}_k(C) \cong \mathcal{S}_{N,k}^{\text{fermion}} \otimes \mathcal{M}_k(C), \quad \dim \mathcal{M}_k(\text{Slater}) = 1. \quad (65)$$

Theorem VII.1 (Cut-rank divisibility obstruction). *For every $N \geq 3$, Eq. (65) cannot hold for every nonzero N -form.*

Proof. Assume Eq. (65) holds for every nonzero N -form. For a Slater determinant S_N , Structural result II at $k = 1$ gives

$$\dim \mathcal{B}_1(S_N) = \text{rank } c_{S_N,1} = \binom{N}{1} = N. \quad (66)$$

Since $\dim \mathcal{M}_1(S_N) = 1$, taking dimensions in the assumed tensor product forces

$$\dim \mathcal{S}_{N,1}^{\text{fermion}} = N. \quad (67)$$

Consequently every admitted state C would satisfy

$$\text{rank } c_{C,1} = \dim \mathcal{B}_1(C) = N \dim \mathcal{M}_1(C), \quad (68)$$

and its one-cut rank would be divisible by N .

Now take $V = \mathbb{F}^{N+2}$ and let

$$C_N = e_1 \wedge (e_2 \wedge e_3 + e_4 \wedge e_5) \wedge e_6 \wedge \cdots \wedge e_{N+2}.$$

Put $w = e_6 \wedge \cdots \wedge e_{N+2}$, with $w = 1$ when $N = 3$. For the dual basis (e_i^*) , contraction gives, up to nonzero signs,

$$\begin{aligned} \iota_{e_1^*} C_N &= (e_2 \wedge e_3 + e_4 \wedge e_5) \wedge w, \\ \iota_{e_2^*} C_N &= e_1 \wedge e_3 \wedge w, \\ \iota_{e_3^*} C_N &= e_1 \wedge e_2 \wedge w, \\ \iota_{e_4^*} C_N &= e_1 \wedge e_5 \wedge w, \\ \iota_{e_5^*} C_N &= e_1 \wedge e_4 \wedge w, \\ \iota_{e_i^*} C_N &= e_1 \wedge (e_2 \wedge e_3 + e_4 \wedge e_5) \\ &\quad \wedge \iota_{e_i^*} w, \quad 6 \leq i \leq N+2. \end{aligned} \quad (69)$$

To prove independence, consider a relation $\sum_{i=1}^{N+2} \gamma_i \iota_{e_i^*} C_N = 0$. A monomial without e_1 occurs only in the $i = 1$ contraction, so $\gamma_1 = 0$. The four

monomials $e_1 \wedge e_3 \wedge w$, $e_1 \wedge e_2 \wedge w$, $e_1 \wedge e_5 \wedge w$, and $e_1 \wedge e_4 \wedge w$ occur only for $i = 2, 3, 4, 5$, respectively, so $\gamma_2 = \cdots = \gamma_5 = 0$. For each $i \geq 6$, the monomial $e_1 \wedge e_2 \wedge e_3 \wedge \iota_{e_i^*} w$ omits a different basis vector from w and occurs in no other remaining contraction, so $\gamma_i = 0$. Therefore

$$\text{rank } c_{C_N,1} = \dim \text{span}\{\iota_{e_i^*} C_N : 1 \leq i \leq N+2\} = N+2. \quad (70)$$

But $N \nmid N+2$ for every $N \geq 3$, contradicting Eq. (68). $\square$

The counterexample is only two Slaters. Shared physical covectors lock $N-2$ channels that an independent multiplicity tensor would treat as free. The rank is invariant under orbital basis changes and direct embeddings, and full one-cut rank is stable on an open neighborhood.

Standard symmetry-adapted tensor networks remain valid [19, 20]. Particle permutation supplies the one-dimensional sign representation, not a $\binom{N}{k}$ degeneracy. Under $GL(V)$, $\Lambda^N V$ is irreducible and the exterior cut has Littlewood–Richardson multiplicity one while retaining the full orbital representation. Hamiltonian-specific charges and multiplets are not ruled out. State-adaptive Slater/secant descriptions are established and need not be canonical [21–23].

The sparse-path construction sharpens the algorithmic warning: whenever its output is nonzero it is projectively one Slater, yet its scale is a permanent. A state-intrinsic multiplicity-one label cannot by itself compile compact cores into a norm algorithm.

VIII. ESTABLISHED POSITIVE CONTROLS AND PRIOR-ART BOUNDARY

Fixed-number AGP is equivalent to number-projected pairing structure; projected BCS, HFB overlaps, and Pfaffian formulas are established [24–26]. AGP norms and reduced density matrices have polynomial algorithms [27]. Deterministic linear combinations of optimized and nonorthogonal AGPs precede this project [28, 29]; Pfaffian variational Monte Carlo with symmetry projection is also established [30, 31].

Accordingly, finite LC-AGP is used here as a validated tractable control, not as a new ansatz. The two collapse theorems above classify when matrix-pair memory reduces to that control. Grassmann signs or parity book-keeping enforce fermionic algebra but do not, without additional closure, refute either hardness construction.

Lemma VIII.1 (Explicit AGP embedding in one-form FEMPS). *Let a scalar two-form be in skew-normal form $\Omega_F = \sum_{a=1}^r u_a u_a \wedge v_a$. For $1 \leq M \leq r$,*

$$\frac{\Omega_F^M}{M!} = \sum_{a_1 < \cdots < a_M} \left(\prod_{j=1}^M w_{a_j} \right) u_{a_1} \wedge v_{a_1} \wedge \cdots \wedge u_{a_M} \wedge v_{a_M}$$

has an exact one-form FEMPS with every internal bond at most $r \leq \lfloor D/2 \rfloor$.

Proof. Put $p_a = u_a \wedge v_a$. Each p_a has even degree, so $p_a \wedge p_b = p_b \wedge p_a$, while $p_a \wedge p_a = 0$. Expanding the power gives

$$\begin{aligned} \Omega_F^M &= \sum_{a_1, \dots, a_M \in [r]} \left(\prod_{j=1}^M w_{a_j} \right) p_{a_1} \wedge \dots \wedge p_{a_M} \\ &= M! \sum_{1 \leq a_1 < \dots < a_M \leq r} \left(\prod_{j=1}^M w_{a_j} \right) p_{a_1} \wedge \dots \wedge p_{a_M}. \end{aligned} \quad (71)$$

The second equality follows because repeated labels vanish and each increasing M -tuple has $M!$ permutations, all with positive graded-commutation sign. Dividing Eq. (71) by $M!$ gives the expansion in the lemma statement.

It remains to realize precisely those increasing tuples as virtual paths. Use the channel set $[r]$. For $M = 1$, take

$$A_{1a}^{[1]} = w_a u_a, \quad A_{a1}^{[2]} = v_a. \quad (72)$$

For $M \geq 2$, define the $2M$ one-form cores by

$$\begin{aligned} A_{1a}^{[1]} &= w_a u_a, \\ A_{ab}^{[2q]} &= \delta_{ab} v_a, & 1 \leq q < M, \\ A_{ab}^{[2q-1]} &= \mathbf{1}_{a < b} w_b u_b, & 2 \leq q \leq M, \\ A_{a1}^{[2M]} &= v_a. \end{aligned} \quad (73)$$

Here every omitted entry is zero. A nonzero contraction of Eq. (73) has a unique channel sequence satisfying

$$1 \leq a_1 < a_2 < \dots < a_M \leq r, \quad (74)$$

and its physical wedge and scalar coefficient are

$$\left(\prod_{j=1}^M w_{a_j} \right) u_{a_1} \wedge v_{a_1} \wedge \dots \wedge u_{a_M} \wedge v_{a_M}. \quad (75)$$

Conversely every increasing sequence produces exactly one such path. Hence the contracted FEMPS equals Eq. (71)/ $M!$. Every internal virtual space is a subspace of $\mathbb{F}^r$, so every internal bond is at most $r \leq \lfloor D/2 \rfloor$. The core description stores the strictly upper-triangular transitions and does not materialize its $\binom{r}{M}$ nonzero paths. $\square$

For odd particle number, a blocked orbital b gives $b \wedge \Omega_F^M/M!$ by prepending one bond-one core; no auxiliary subtraction is needed for this representation statement. This embedding is a prior-art control, not a new AGP ansatz. It also does not contradict the $O(\chi D)$ bond that can arise when a general matrix-valued pair core of bond χ is split into one-form sites: skew-normal scalar AGP structure is the special closure used by the lemma.

IX. NUMERICAL VALIDATION AND ALGORITHM-DESIGN CONSEQUENCES

The structural results were subsequently used to choose, rather than reject, a restricted numerical control. A nonbranching diagonal-path subclass is exactly a K -term nonorthogonal Slater expansion and evaluates determinant transition quantities in $O(K^2)$ path pairs without enumerating virtual histories or materializing the alternating coefficient tensor. This places it within established NOCI methodology rather than defining a new correlated ansatz [32, 33].

We retain one representative interacting result. For six one-dimensional fermions with harmonic confinement and softened Coulomb repulsion, the $D = 12$, $K = 4$ calculation occupies the $\binom{12}{6} = 924$ dimensional exterior space and gives energy 25.0494711446, compared with same-basis CI 25.0493664161. The error is 1.0473×10^{-4} , the energy variance is 1.1223×10^{-3} , the norm error is 1.55×10^{-15} , and the structural antisymmetry residual is exactly zero. The run took 107.7 s on CPU and its sampled peak resident memory was 0.975 GB. The $D = 12$, $K = 4$ orbitals were *not* initialized from a clean $D = 12$ Slater: they were obtained by exact zero-padding of a preoptimized $D = 10$, $K = 4$ state before the $D = 12$ optimization. Direct CI was used only as an independent reference.

These floating-point data are a bounded NOCI-equivalent numerical exercise. They do not establish an ansatz beyond NOCI, an explicit-correlation advantage, improved functional-basis D convergence, asymptotic scalability, or superiority over FCI, particle TT, Li-Waintal, or same-basis DMRG.

X. DISCUSSION

A. Scope of the complexity statements

The exact reductions and approximation statements answer different questions. Exact #P-hardness does not rule out an FPRAS, an additive estimator, or an efficient algorithm under a promise that excludes the hard instances. Norm polarization recovers an affine signed scale, whereas the sparse-path output is projectively one Slater; a simple normalized state need not be easy to normalize from compact cores. All input-complexity statements count the explicit rational core data and coefficient bit length, not only the intrinsic Slater rank of the represented state.

The representation boundary is equally important. The exchange-rank results concern labelled-particle TT, while the fixed-bond hardness result concerns exterior one-form cores. Neither transfers automatically to mode MPS, ordered sectors, or every fermionic tensor network. Likewise, the matrix-pair collapse repeats one homogeneous even form, whereas the Cayley construction uses site-labelled one-form cores whose products retain row

order. Finally, controlled energy estimation requires simultaneous numerator and norm bounds with a strictly positive lower norm bound; separate unbiased estimators do not by themselves certify their ratio.

B. Two mechanisms in structured antisymmetrization

Consider exact evaluation at a fixed configuration $x = (x_1, \dots, x_N)$ of

$$(\mathcal{A}F)(x) = \sum_{\pi \in S_N} \text{sgn}(\pi) F(x_{\pi(1)}, \dots, x_{\pi(N)}). \quad (76)$$

Two independent properties organize several familiar constructions. Property P1 (*assignment factorization*) asks whether, after fixing x , every permutation term has the form

$$F_\pi(x) = \prod_{i=1}^N a_{i, \pi(i)}(x), \quad (77)$$

where the permutation selects factors only through $(i, \pi(i))$. The factors $a_{ij}(x)$ may depend on the *entire* configuration, as in backflow or FermiNet; P1 is not a one-particle locality assumption. Property P2 asks whether these factors multiply in one commutative coefficient algebra. In every cell below we additionally assume that each factor is itself evaluable from its encoded input in polynomial time and polynomial bit complexity. Without this base assumption, an easy determinant layer need not make the complete model easy.

In the upper-left cell, the ordinary Slater determinant replaces an explicit $N!$ -term Leibniz sum by one determinant. At fixed x , generalized Slater, backflow, and FermiNet orbital entries remain scalars in a commutative field, even when every entry depends on all electronic coordinates; permutation equivariance makes particle exchange act by row or column exchange. Each determinant channel is therefore polynomial-time once its entries are available. Gaussian elimination applies over fields. Over a general commutative ring, divisions need not exist, but division-free determinant algorithms such as Berkowitz's still give polynomial-size arithmetic circuits [34]. Hutter's generalized-Slater universality analysis, FermiNet, the explicitly antisymmetrized layers of Lin–Goldshlager–Lin, and the separation of Zweig–Bruna concern representation, universality, ansatz design, or expressivity [35–38]; they are not attributed the complexity map in Table I.

The upper-right cell isolates the second mechanism. In one-form FEMPS, Eq. (77) holds with matrix-valued factors, but row order prevents their exchange. The exterior algebra supplies $\text{sgn}(\pi)$ while the virtual algebra retains the noncommuting product, producing the row-ordered Cayley determinant rather than a scalar determinant. The following point-evaluation theorem makes

this distinction explicit without changing the squared-norm theorem above.

Theorem X.1 (Exact rational-polynomial point evaluation). *Let $\ell_r(t) = P_r(2t - 1)$, $r \geq 0$, be the unnormalized shifted Legendre polynomials, where $P_r(1) = 1$. Exact pointwise evaluation of rational one-form FEMPS represented in the basis $\ell_0, \dots, \ell_{D-1}$ is $\#P$ -hard under polynomial-time metric reductions, already when $D = N$ and every internal bond is at most two. For the isometric convention J_N , the exact output is encoded as the rational q in $\Psi(x) = q/\sqrt{N!}$; equivalently one may request the unnormalized rational alternating value q .*

Proof. Functional basis and evaluation matrix. The shifted polynomials have the integer expansion

$$\ell_r(t) = \sum_{k=0}^r (-1)^{r-k} \binom{r}{k} \binom{r+k}{k} t^k \quad (78)$$

and leading coefficient $c_r = \binom{2r}{r}$. Put

$$t_i = \frac{i}{N+1} \quad (i \in [N]), \quad B_{ij} = \ell_{j-1}(t_i) \quad (i, j \in [N]). \quad (79)$$

Writing the polynomials in increasing degree expresses $B = VT$, where $V_{i,k+1} = t_i^k$ is the monomial Vandermonde matrix and T is upper triangular with diagonal $(c_0, \dots, c_{N-1})$. Therefore

$$\det B = \left(\prod_{r=0}^{N-1} \binom{2r}{r} \right) \prod_{1 \leq i < k \leq N} (t_k - t_i) \neq 0. \quad (80)$$

The nodes have $O(\log N)$ -bit numerator and denominator. Since $r, k \leq N-1$, every binomial coefficient in Eq. (78) has $O(N)$ bits. Let

$$Q = (N+1)^{N-1}, \quad A := QB. \quad (81)$$

For $j-1 \leq N-1$, the denominator of $\ell_{j-1}(i/(N+1))$ divides $(N+1)^{j-1}$, so A is an integer matrix. Each A_{ij} is a sum of at most N integers, each with $O(N \log N)$ bits; hence

$$\text{bits}(A_{ij}) = O(N \log N). \quad (82)$$

If $A^{(r,s)}$ denotes any $(N-1) \times (N-1)$ minor, Hadamard's inequality gives

$$\begin{aligned} \log_2 |\det A| &\leq N(O(N \log N) + \tfrac{1}{2} \log_2 N), \\ \log_2 |\det A^{(r,s)}| &\leq (N-1)(O(N \log N) + \tfrac{1}{2} \log_2 N). \end{aligned} \quad (83)$$

Thus $\det B = \det A/Q^N$, its reciprocal, and

$$B^{-1} = QA^{-1} = Q \frac{\text{adj}(A)}{\det A} \quad (84)$$

all have polynomial bit length and are computable by exact rational Gaussian elimination in polynomial time.

	P2: commutative	not P2: noncommutative
P1	Slater, generalized Slater, backflow, and FermiNet determinant channels: the $N!$ Leibniz terms collapse to ordinary determinants.	One-form FEMPS: exterior signs accompany row-ordered virtual products, giving a Cayley determinant.
not P1	The signed Hamilton-path polynomial in Eq. (94): open cell in the present bounded search.	General explicit antisymmetrization; conditional hardness only for a precisely defined class that exactly contains the hard bond-two one-form cores with polynomial overhead.

TABLE I. A two-mechanism map, not a complete dichotomy or classification theorem. P1 concerns assignment structure; P2 concerns the coefficient algebra.

Source instance. Let φ be a 3CNF formula with m clauses and $S = \#\text{SAT}(\varphi)$. Take the CHSS matrix $H_\varphi = (H_{ij})_{i,j \in [N]} \in \text{Mat}_N(\text{Mat}_2(\mathbb{Q}))$ used in the proof of Theorem IV.2, where $N = \text{poly}(|\varphi|)$ and

$$u^\top \text{CDet}(H_\varphi) v = 4^{3m} S, \quad u = e_1^{(2)}, \quad v = e_1^{(2)} + e_2^{(2)}. \quad (85)$$

AUDIT FLAG. Equation (85) imports the same structured CHSS theorem as the squared-norm reduction, including row ordering, characteristic-zero validity, polynomial construction size, and gadget coefficient bounds. This external lemma is not reproved here and requires independent source verification. The standard $\#\text{P}$ -completeness of $\#\text{SAT}$ is also used as the source-problem classification rather than reproved here.

Constructed functional FEMPS instance. Set $D = N$ and define

$$\mathcal{A}^{[i]}(t) = \sum_{j=1}^N H_{ij} \ell_{j-1}(t), \quad u = e_1, \quad v = e_1 + e_2. \quad (86)$$

Absorbing u, v into the endpoints gives maximum internal bond two. The functional-basis coefficients are exactly the four rational entries of the N^2 matrices H_{ij} , so their number and bit length are polynomial in $|\varphi|$.

Oracle value. With the isometric alternating convention, direct point evaluation is

$$\begin{aligned} \Psi(t_1, \dots, t_N) &= \frac{1}{\sqrt{N!}} \sum_{\pi \in S_N} \text{sgn}(\pi) u^\top \prod_{i=1}^N \mathcal{A}^{[i]}(t_{\pi(i)}) v \\ &= \frac{1}{\sqrt{N!}} \sum_{j_1, \dots, j_N \in [N]} u^\top H_{1j_1} \cdots H_{Nj_N} v \\ &\quad \times \Delta(j_1, \dots, j_N), \end{aligned} \quad (87)$$

where

$$\begin{aligned} \Delta(j_1, \dots, j_N) &:= \sum_{\pi \in S_N} \text{sgn}(\pi) \prod_{i=1}^N B_{\pi(i), j_i} \\ &= \det[B_{p, j_i}]_{p, i \in [N]}. \end{aligned} \quad (88)$$

If two j_i coincide, this determinant has repeated columns and vanishes. If $j_i = \sigma(i)$ for $\sigma \in S_N$, then the matrix

is obtained from B by the column permutation σ , and therefore

$$\Delta(\sigma(1), \dots, \sigma(N)) = \text{sgn}(\sigma) \det B. \quad (89)$$

Equation (87) consequently becomes

$$\begin{aligned} \Psi(t_1, \dots, t_N) &= \frac{\det B}{\sqrt{N!}} \sum_{\sigma \in S_N} \text{sgn}(\sigma) u^\top H_{1\sigma(1)} \cdots H_{N\sigma(N)} v \\ &= \frac{\det B}{\sqrt{N!}} u^\top \text{CDet}(H_\varphi) v \\ &= \frac{\det B 4^{3m} S}{\sqrt{N!}}. \end{aligned} \quad (90)$$

Source answer and output encoding. The exact oracle returns the pair (q, N) representing $q/\sqrt{N!}$, where

$$q = \det B 4^{3m} S \in \mathbb{Q}. \quad (91)$$

Since $\det B \neq 0$, the source answer is

$$S = \frac{q}{4^{3m} \det B}. \quad (92)$$

The integer $N!$ has $O(N \log N)$ bits; $S \leq 2^{n_{\text{var}}}$; and the preceding determinant bounds show that q , $\det B$, and $(\det B)^{-1}$ have polynomial bit length. Exact rational division in Eq. (92) is polynomial-time. Thus

$$\varphi \mapsto (\mathcal{A}^{[1]}, \dots, \mathcal{A}^{[N]}; t_1, \dots, t_N) \mapsto (q, N) \mapsto S \quad (93)$$

is a polynomial-time metric reduction. For the unnormalized antisymmetrizer, the same chain omits the known factor $1/\sqrt{N!}$. $\square$

Theorem X.1 and Theorem IV.2 are independent reductions from the same Cayley source; neither is inferred from the other. They concern worst-case exact evaluation and do not preclude efficient approximate QMC/VMC, promised subclasses, or particular low-bond instances.

For the lower-left cell, the exact path object is

$$H_{\text{sgn}}(W) = \sum_{\pi \in S_N} \text{sgn}(\pi) \prod_{i=1}^{N-1} w_{\pi(i), \pi(i+1)}. \quad (94)$$

Its scalar factors commute, but adjacent vertices couple successive permutation positions, so it need not have

P1. This is a path, not a closed cycle: summing a cycle over all permutations would cancel for even N because a cyclic shift reverses permutation parity. Hamilton path/cycle polynomials, sign imbalance, even-minus-odd path counts, fermionants, and immanants were checked in a bounded search; the latter neighboring families have their own complexity results [39]. In the present bounded literature search, the authors did not find an existing complexity classification for this exact weighted polynomial. We therefore register the cell as open and make no hardness or tractability claim.

The lower-right entry supports only a conditional inheritance statement. If a precisely defined parameterized function class exactly represents every hard bond-two instance in Theorem X.1 with polynomial overhead, then its exact point-evaluation problem inherits that #P-hardness. Universal approximation does not imply exact containment. In particular, Lin–Goldshlager–Lin motivate generic explicit antisymmetrization, including its factorial sum, but no exact containment lemma is asserted here for their neural class.

Finally, if J is fully permutation-symmetric, then directly from Eq. (76),

$$J\mathcal{A}(M) = \mathcal{A}(JM). \quad (95)$$

If J has a compatible scalar particle-TT representation of bond χ_J and multiplication closes in the chosen functional basis or in a stated finite extension, tensoring its virtual indices with a Slater carrier gives a conditional Slater–Jastrow embedding with the corresponding χ_J bond factor. This is an embedding statement, not a norm-hardness reduction; the point value remains $J(x)\det\Phi(x)$ up to normalization. The Zweig–Bruna separation concerns expressive efficiency relative to finite Slater sums, not point- or norm-evaluation complexity.

The table is an organizing framework and partial classification program. A complete complexity classification of structured antisymmetrization, including the unresolved cells and internal tractability boundaries, is left for future work.

C. Limitations and next questions

The results do not establish that every exterior family is hard, that every tractable exterior family is LC–AGP, or that additive approximation with a certified polynomial norm lower bound is impossible. Promised positive, low-rank, strongly orthogonal, Pfaffian/Gaussian, integrable, or low-treewidth families remain admissible if their promise excludes the hard embeddings and their state and energy errors are certified. A genuinely different categorical carrier is not excluded by the divisibility obstruction; it must state its exactness, reconstruction cost, and contraction theorem rather than assume a free direct product.

The structural results and all currently admitted restricted-solver numerics belong to this single

manuscript. No separate FEMPS method paper is claimed from the diagonal-path data. A future method manuscript is opened only after a representation beyond finite NOCI demonstrates a matched, reproducible advantage—for example, explicit correlation that improves functional-basis D convergence, or a controlled comparison with Li–Waintal and same-basis DMRG that identifies a genuine accuracy, stability, or complexity tradeoff. Higher-dimensional alternating-form rank classification is not pursued unless a specific rank question determines such an algorithmic or physical decision.

XI. CONCLUSION

The attempted fermionic completion of the 2201 functional tensor network has identified two costs that must not be conflated. Ordinary particle TT pays a binomial exchange bond even for a Slater determinant. Exteriorizing the state removes that representation charge, but generic exact squared-norm contraction and rational-polynomial point evaluation are already hard at fixed one-form bond, while a direct universal carrier factorization fails dimensionally. The tested tractable matrix-pair memories reduce to established LC–AGP structure, and a unique sparse path already contains a permanent. The nonbranching K -path calculation is correspondingly a useful NOCI-equivalent numerical control, not yet an independent method contribution. The resulting map is an algorithm-design constraint, not a denial of numerical FEMPS: genuinely distinct restricted subclasses and error-controlled approximate methods remain open and must be judged by matched interacting benchmarks, D and correlation convergence, cost, uncertainty, and antisymmetry residuals rather than by parameter counts alone.

USE OF ARTIFICIAL-INTELLIGENCE TOOLS

OpenAI Codex (GPT-5 family, accessed in August–September 2026) assisted with literature discovery, code and proof exploration, numerical cross-checks, and language editing under human direction. The human author selected the research questions, inspected the cited sources, verified the mathematical and numerical results, revised the manuscript, and accepts responsibility for its content. The AI system is not an author.

Appendix A: Detailed bounded-algebra collapse proof

This appendix expands the algebraic steps used in the bounded-algebra collapse theorem. Let $E = \Lambda^{\text{even}}V_D$. The wedge product makes E a commutative $\mathbb{C}$ -algebra, although it has zero divisors. Write $\Omega = \Omega_S + \Omega_R \in \mathcal{A} \otimes E_2$, where $E_2 = \Lambda^2V_D$.

a. Radical filtration. Because R is a two-sided ideal, a product containing k factors from R , with arbitrary elements of S between them, belongs to R^k . This follows inductively: $SR^j, R^jS \subseteq R^j$, and $R^iR^j \subseteq R^{i+j}$. Hence every word in the expansion of Ω^M with $k \geq d$ radical factors vanishes. For fixed $k < d$, every surviving word has a unique run-length description

$$\Omega_S^{m_0} \Omega_R \Omega_S^{m_1} \cdots \Omega_R \Omega_S^{m_k}, \quad m_0 + \cdots + m_k = M - k,$$

where zero-length semisimple runs are allowed. Summing over all nonnegative run lengths is equivalent to retaining the homogeneous degree- M part of the commutative physical-form polynomial constructed below; no additional combinatorial factor is omitted because the virtual product order remains the displayed order.

b. Wedderburn blocks and radical basis. Let 1_a be the central identity of the block $S_a \cong \text{Mat}_{p_a}(\mathbb{C})$ and choose a basis $r_1, \dots, r_\rho$ of R . Insert $1_S = \sum_{a=1}^q 1_a$ independently into each of the $k+1$ semisimple runs and expand each radical factor as

$$\Omega_R = \sum_{b=1}^{\rho} r_b \eta_b, \quad \eta_b \in E_2.$$

Thus the contribution with k radical occurrences is a sum over at most $q^{k+1}\rho^k$ choices $\tau = (a_0, b_1, a_1, \dots, b_k, a_k)$. This is an upper bound: incompatible block transitions may vanish and repeated basis choices may combine.

Fix one τ . In matrix units of S_{a_j} , the corresponding projected semisimple form has at most $p_{a_j}^2 \leq p^2$ scalar two-form entries $x_{j,\mu\nu} \in E_2$. Together with the selected radical forms $\eta_{b_1}, \dots, \eta_{b_k}$, the boundary value under λ is a homogeneous degree- M polynomial

$$P_\tau(x_{0,\mu\nu}, \dots, x_{k,\mu\nu}, \eta_{b_1}, \dots, \eta_{b_k}) \in E_{2M}$$

in at most $v_k = \sum_{j=0}^k p_{a_j}^2 + k \leq (k+1)p^2 + k$ commuting two-forms. Matrix multiplication and λ affect only the scalar coefficients of this polynomial. Nilpotence inside E can identify or kill monomials, but cannot increase the number of powers needed below.

c. Polarization into AGPs. Let $z_1, \dots, z_v$ be formal commuting variables. Over characteristic zero, put $T_M = \{t \in \mathbb{Z}_{\geq 0}^{v-1} : |t| \leq M\}$. The integer simplex is unisolvant for polynomials of total degree at most M : ordered by total degree, the falling-factorial polynomials $\prod_{i=2}^v \binom{x_i}{\alpha_i}$ give a triangular evaluation matrix on T_M with diagonal one. Hence the square matrix relating the $|T_M| = \binom{M+v-1}{v-1}$ powers

$$(z_1 + t_2 z_2 + \cdots + t_v z_v)^M, \quad t \in T_M,$$

to the degree- M monomial basis is nonsingular (the multinomial factors are nonzero in characteristic zero). Therefore every homogeneous degree- M polynomial in v variables is a linear combination of at most $|T_M|$ such powers.

For rational data this is also a Turing construction. The evaluation entries are integers of bit length $O(M \log(M+1))$; for fixed v the matrix order is $O(M^{v-1})$. Hadamard's inequality and Cramer's rule bound every numerator and denominator in its inverse by a polynomial number of bits. Exact Gaussian elimination therefore computes the power coefficients in polynomial time. In the theorem, $v \leq v_k$ is fixed when p, d are fixed; the supplied rational block decomposition and radical basis have polynomial-bit entries, and the number of structural choices is polynomial. Multiplication, application of λ , and division by $M!$ increase bit lengths only polynomially.

Substitute the scalar two-forms for the z_i and apply the algebra homomorphism from the symmetric algebra to E . Every resulting term $L^M/M!$ is an AGP. For a fixed k , multiplying the power count by the at most $q^{k+1}\rho^k$ structural choices and summing over $0 \leq k \leq \min(d-1, M)$ gives Eq. (31). All steps use only the fixed Wedderburn decomposition, radical basis, matrix multiplication, and exact linear algebra. For fixed p and d , the displayed count is polynomial in M, q , and ρ , hence in the explicit algebraic input size.

Appendix B: Detailed fixed-state graded collapse proof

Let $R_d = \mathbb{Q}[z_1, \dots, z_g]/(z_1^{d_1}, \dots, z_g^{d_g})$ and suppose the embedding into $\text{Mat}_w(R_d)$ is part of the input. Represent every residue by its canonical polynomial lift with $0 \leq \deg_{z_j} < d_j$. An entry of the unreduced product Ω^M then has $\deg_{z_j} \leq L_j = M(d_j - 1)$. For each counter use the rational grid $0, 1, \dots, L_j$. The univariate Vandermonde matrix is nonsingular, and the tensor product of these matrices recovers every coefficient of the lifted product. Reducing modulo $z_j^{d_j}$ simply discards coefficients whose z_j exponent is at least d_j ; it cannot create additional terms.

At one grid point, Ω is a $w \times w$ matrix whose w^2 entries are commuting physical two-forms. After the fixed boundary is applied, Ω^M is a homogeneous degree- M polynomial in at most w^2 such forms. The integer-simplex polarization construction in Appendix A expresses it as at most $\binom{M+w^2-1}{w^2-1}$ powers. Recombining the at most $\prod_j (L_j + 1)$ grid contributions gives Eq. (V.2).

For the bit model, every Vandermonde entry has polynomial bit length in L_j ; Hadamard's inequality and Cramer's rule give polynomial bit length for its inverse. Because g is fixed and the quotient basis of dimension $\prod_j d_j$ is explicit, $\prod_j (L_j + 1)$ is polynomial in the stated input and M . The polarization dimension w^2 is fixed. Exact rational matrix multiplication, tensor-product interpolation, quotient reduction, boundary application, and division by $M!$ therefore use polynomially many bit operations and produce polynomial-bit LC-AGP coefficients. This conclusion requires the rational embedding

and quotient representation as input; it does not claim

an algorithm for discovering such an embedding from arbitrary structure constants.

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